An explicit finite-kinase instability bound for a mixed-mechanism phosphorylation system

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Abstract

For the mass-action mixed-mechanism phosphorylation system of Conradi, Mincheva and Shiu, at their Table 1 rates and fixed phosphatase and substrate totals 8 and 40, we prove that the unique positive equilibrium has exactly two unstable and four stable compatibility-class eigenvalues whenever the kinase total is at least $4660166/209569 < 23$. A rational equilibrium parameterization converts the unbounded range of kinase totals into a bounded interval. Exact Bernstein certificates for six Hurwitz determinants then give a uniform Routh root count. The bound is sufficient; sharpness is not asserted. This proves the first assertion of their Conjecture 6.2 with explicit witness totals. Its periodic-orbit assertion, and hence the full original problem, remain unresolved here.

1 Model, statement and relation to prior work

Consider processive phosphorylation and distributive dephosphorylation:

$$S_0 + K \underset{k_2}{\overset{k_1}{\rightleftarrows}} S_0K \xrightarrow{k_3} S_1K \xrightarrow{k_4} S_2 + K,$$
$$S_2 + P \underset{k_6}{\overset{k_5}{\rightleftarrows}} S_2P \xrightarrow{k_7} S_1 + P \underset{k_9}{\overset{k_8}{\rightleftarrows}} S_1P \xrightarrow{k_{10}} S_0 + P.$$

Use the species order

$$(x_1, \dots, x_9) = (S_0, K, S_0K, S_1K, S_2, P, S_2P, S_1, S_1P)$$

and the fixed rate vector

$$(k_1, \dots, k_{10}) = (1, 1, 1, 1, 100, 1, 9/10, 3, 1, 100). \tag{1}$$

Thus, in the notation of Conradi–Mincheva–Shiu [1], the equations are

$$\begin{aligned}
\dot{x}_1 &= -x_1x_2 + x_3 + 100x_9, & \dot{x}_2 &= -x_1x_2 + x_3 + x_4,\\
\dot{x}_3 &= x_1x_2 - 2x_3, & \dot{x}_4 &= x_3 - x_4,\\
\dot{x}_5 &= x_4 - 100x_5x_6 + x_7, & \dot{x}_7 &= 100x_5x_6 - \tfrac{19}{10}x_7,\\
\dot{x}_6 &= -100x_5x_6 - 3x_8x_6 + \tfrac{19}{10}x_7 + 101x_9, & &\\
\dot{x}_8 &= \tfrac{9}{10}x_7 - 3x_6x_8 + x_9, & \dot{x}_9 &= 3x_6x_8 - 101x_9.
\end{aligned} \tag{2}$$

The conserved totals are

$$\kappa = x_2 + x_3 + x_4, \qquad P_{\mathrm{tot}} = x_6 + x_7 + x_9, \qquad S_{\mathrm{tot}} = x_1 + x_3 + x_4 + x_5 + x_7 + x_8 + x_9. \tag{3}$$

Here κ denotes the total kinase, to distinguish it from free kinase. The three independent conservation laws define a six-dimensional compatibility class in the nonnegative orthant.

Theorem 1. *For* (2) *with* $P_{\mathrm{tot}} = 8$ *and* $S_{\mathrm{tot}} = 40$*, and every finite*

$$\kappa \geq B := \frac{4660166}{209569},$$

the unique positive equilibrium has exactly two eigenvalues in $\{\lambda : \operatorname{Re}\lambda > 0\}$ *and four in* $\{\lambda : \operatorname{Re}\lambda < 0\}$ *on its compatibility class, counted with algebraic multiplicity. In particular, this holds for every* $\kappa \geq 23$*.*

The source paper [1], especially Proposition 3.1, Proposition 4.1 and Section 6, provides equilibrium uniqueness, a stability-surface analysis and large-kinase simulations. Its Conjecture 6.2(1) asks for fixed phosphatase and substrate totals admitting unstable equilibria at arbitrarily large kinase totals. Theorem 1 supplies these witnesses and certifies an entire unbounded ray, excluding restabilization on that ray. Equilibrium parameterization and the Hurwitz method themselves are established tools.

There is direct subsequent topical overlap: Grasse's 2023 poster-programme entry concerns kinase-total robustness [2], and the 2025 abstract of Grasse and Conradi considers the reversed processive/distributive arrangement [3]. The inspected programme and abstract give no equivalent finite bound or complete proof. The poster itself and possible follow-up texts were unavailable to this comparison. No priority claim is made. The contribution here is the explicit bound and exact certificate for the stated fixed totals, rather than the prior observation of oscillations.

2 Equilibrium parameter and coverage of the kinase ray

Put $z = x_9$ and define

$$\begin{aligned} p(z) &= 8 - \frac{1009}{9}z, \quad G(z) = \frac{2809}{9}z + \frac{322z}{9p(z)}, \\ r(z) &= 40 - G(z), \quad \kappa(z) = 200z + \frac{200z}{r(z)}. \end{aligned} \tag{4}$$

At a positive equilibrium, direct balance in (2) yields

$$x^*(z) = \left(r, \frac{200z}{r}, 100z, 100z, \frac{19z}{9p}, p, \frac{1000z}{9}, \frac{101z}{3p}, z\right). \tag{5}$$

For example, $\dot{x}_4 = \dot{x}_1 = \dot{x}_3 = 0$ gives $x_4 = x_3 = 100z$ and $x_1x_2 = 200z$. The x_5, x_7 balances give $x_7 = 1000z/9$ and $x_5 = 19z/(9p)$; the x_9 balance gives $x_8 = 101z/(3p)$. The phosphatase and substrate totals then give p and r. Conversely, substitution of (5) into all nine equations and (3) verifies all these identities.

On $0 < z < 72/1009$ we have $p > 0$ and

$$G'(z) = \frac{2809}{9} + \frac{2576}{9p(z)^2} > 0.$$

Since $G(0) = 0$ and $G(z) \to \infty$ at the right endpoint, there is exactly one α in this interval satisfying $G(\alpha) = 40$. The positive equilibria correspond precisely to $0 < z < \alpha$. On this interval,

$$\kappa'(z) = 200 + \frac{200}{r(z)} + \frac{200zG'(z)}{r(z)^2} > 0.$$

The endpoint limits of $\kappa(z)$ are 0 and ∞. Thus (5) gives exactly one positive equilibrium for every positive finite total kinase. No interval of totals is omitted.

For an explicit enclosure define

$$Q(z) = 2834281z^2 - 568386z + 25920, \qquad L = \frac{7}{100}, \qquad U = \frac{70122383355}{10^{12}}. \tag{6}$$

Then $r = Q/[9(72 - 1009z)]$, and exact evaluation gives

$$Q(L) = \frac{209569}{10000} > 0, \qquad Q(U) = -\frac{6175117557936279}{4000000000000000000000} < 0.$$

Moreover $0 < L < U < 72/1009$ and $Q'(U) < 0$. Because $Q'' > 0$, the polynomial is strictly decreasing on $[0, U]$. Consequently $L < \alpha < U$, and $Q > 0$ on $[L, \alpha)$. Finally $\kappa(L) = B < 23$. Strict monotonicity proves that $[L, \alpha)$ maps bijectively onto $[B, \infty)$.

3 Exact signs and eigenvalue count

In the compatibility class retain coordinates $(x_3, x_4, x_5, x_7, x_8, x_9)$ and eliminate

$$x_1 = 40 - x_3 - x_4 - x_5 - x_7 - x_8 - x_9, \quad x_2 = \kappa - x_3 - x_4, \quad x_6 = 8 - x_7 - x_9.$$

Differentiate at fixed κ, then substitute (5) and $\kappa = \kappa(z)$. With $u = 200z/r$, $w = 19z/(9p)$ and $v = 101z/(3p)$, the resulting Jacobian is

$$J(z) = \begin{pmatrix} -u - r - 2 & -u - r & -u & -u & -u & -u \\ 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 1 & -100p & 100w + 1 & 0 & 100w \\ 0 & 0 & 100p & -100w - 19/10 & 0 & -100w \\ 0 & 0 & 0 & 9/10 + 3v & -3p & 1 + 3v \\ 0 & 0 & 0 & -3v & 3p & -3v - 101 \end{pmatrix}. \tag{7}$$

This is a restriction to the tangent space of the conservation laws, so its six eigenvalues exclude the three conservation zeros of the full system. In particular, $\kappa(z)$ is not differentiated as a function of state.

Write

$$\det(\lambda I - J(z)) = \sum_{i=0}^{6} c_i(z)\lambda^{6-i}, \qquad c_0 = 1, \qquad \Delta_j(z) = \det\left[c_{2b-a+1}(z)\right]_{a,b=0}^{j-1},$$

where $c_i = 0$ outside $0 \le i \le 6$. This is the usual Hurwitz convention; its transpose gives the same leading determinants. Let $A = 1009z - 72$ and use the denominators D_j in Table 1. Define the polynomials $N_j = D_j\Delta_j$ and

$$h_j = -N_j \quad (1 \le j \le 5), \qquad h_6 = N_6.$$

The displayed Jacobian and these definitions specify every polynomial without the need for expanded, large integer expressions.

Lemma 1. *For all $z \in [L, \alpha)$,*

$$(\operatorname{sgn}\Delta_1, \ldots, \operatorname{sgn}\Delta_6) = (+, +, +, +, -, -).$$

Proof. For a polynomial h of degree n, put

$$a_\ell = \frac{h^{(\ell)}(L)}{\ell!}(U-L)^\ell, \qquad b_k = \sum_{\ell=0}^{k} a_\ell \frac{\binom{k}{\ell}}{\binom{n}{\ell}}.$$

The exact change of basis is

$$h(L+(U-L)t) = \sum_{k=0}^{n} b_k \binom{n}{k} t^k (1-t)^{n-k}. \tag{8}$$

The basis functions on the right are nonnegative and sum to one on $[0,1]$. Strict positivity of every b_k therefore proves strict positivity of h on the closed interval $[L,U]$.

Exact rational arithmetic applied to (7) gives Table 1. Its last two columns are conservative rational lower bounds, not rounded estimates: for each row, $b_0 = h_j(L) > 10^{m_j}$ and $b_k/b_0 > \eta_j$ for every $0 \le k \le n_j$. Thus all $5+11+18+25+31+35 = 125$ Bernstein coefficients are strictly positive. The supplementary verifier reconstructs these calculations from (7); expanded polynomials and all rational coefficients accompany the note.

j	D_j	$\operatorname{sgn} D_j$	n_j	m_j	η_j
1	$90AQ$	$-$	4	5	24/1000
2	$72900A^3Q^2$	$-$	10	14	28/10000
3	$177147000A^5Q^3$	$-$	17	24	14/10000
4	$71744535000A^7Q^4$	$-$	24	34	8/10000
5	$871696100250A^8Q^5$	$+$	30	41	4/10000
6	$94143178827A^7Q^6$	$-$	34	47	4/10000

Table 1: Exact sign certificate. Denominator signs are used only on $[L,\alpha)$; numerator positivity is certified on $[L,U]$. The exponent m_j means $b_0 > 10^{m_j}$, and η_j is a strict lower bound for every b_k/b_0.

On the physical interval, $A<0$ and $Q>0$, so the denominator signs in the table follow directly. The signs of $N_1,\dots,N_6$ are $(-,-,-,-,-,+)$ by (8). Division gives the claimed signs. Only the numerator polynomials are certified beyond α: no determinant is evaluated across its pole there. □

Proof of Theorem 1. The Routh array is regular because every leading Hurwitz determinant is nonzero. Its first column is

$$1, \quad \Delta_1, \quad \frac{\Delta_2}{\Delta_1}, \quad \frac{\Delta_3}{\Delta_2}, \quad \frac{\Delta_4}{\Delta_3}, \quad \frac{\Delta_5}{\Delta_4}, \quad \frac{\Delta_6}{\Delta_5}.$$

By Lemma 1, its signs are $(+,+,+,+,+,-,+)$, with exactly two sign changes. The classical Routh root-count theorem gives two roots in the open right half-plane, counted with algebraic multiplicity. There are no imaginary-axis roots: $c_6 = 0$ would force $\Delta_6 = 0$; a nonzero imaginary conjugate pair would give a common root of the even and odd parts of the characteristic polynomial, forcing $\Delta_5 = 0$. Both alternatives are excluded. Thus the other four roots lie in the open left half-plane. The bijection $[L,\alpha) \to [B,\infty)$ proves the result for every total stated in the theorem, including $\kappa = B$. □

4 Reproducibility and limitations

The supplementary file `verify_certificate.py` constructs (7), computes the characteristic polynomial and the six Hurwitz determinants over rational functions, and checks every bound in Table 1. Specifically, it checks the displayed denominator factorizations and degrees, computes the derivatives defining a_ℓ, forms every b_k, and compares them with the exact rational lower bounds. It also checks its reconstructed characteristic polynomial and all 125 coefficients against the supplied full certificates. All sign decisions are exact; no integration, fitted asymptotic law, floating-point eigenvalue calculation or exchange of limits is used. The calculations were performed with Python and SymPy 1.13.3. A separate verification pass reconstructed the published nine equations in a different affine coordinate chart and checked the complete polynomial identities using rational arithmetic, including a standard-library `Fraction` expansion of the Bernstein certificates.

The theorem is restricted to (1) and the fixed totals $8, 40$. The threshold is sufficient and is not asserted to be the stability boundary. An unstable equilibrium in six dimensions does not by itself establish a periodic orbit. Neither periodic-orbit existence for all sufficiently large kinase totals nor invariance of its S_2 projection is proved here. These are the additional requirements of Conjecture 6.2(2) in [1]; the full original problem remains unresolved by this note.

Computational and AI assistance. OpenAI Codex assisted the symbolic derivation, exact-verification code, literature screening, drafting and editing. A separate model context checked the frozen mathematical argument and independently reconstructed its algebra. This assistance and the executable checks are not formal proof-assistant verification, human peer review or a guarantee of priority. The mathematical argument and exact certificate specify the claims supported here.

References

[1] C. Conradi, M. Mincheva and A. Shiu, Emergence of oscillations in a mixed-mechanism phosphorylation system, *Bulletin of Mathematical Biology* **81** (2019), 1829–1852. doi:10.1007/s11538-019-00580-6. Model and conjecture numbering follow arXiv:1809.02886v2.

[2] K.-J. Grasse, Analysis of the robustness of oscillations in a mixed mechanism double phosphorylation system with respect to total kinase concentrations, *Systems Biology of Human Disease 2023*, poster programme, poster 48, p. 52 (2023). Official poster programme. Programme entry only; the poster text was not inspected.

[3] K.-J. Grasse and C. Conradi, On the robustness of oscillations in a mixed mechanism phosphorylation system against perturbations in the total amount of the processive enzyme, *BIOMATH 2025 Conference Abstracts*, Sofia, 15–20 June 2025. Official abstract. Conference abstract.